# Assignment 3: Instrumental Variable Estimation

Replicating the IV strategy in Stokes (2015)

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## Assignment instructions:

Working with classmates to troubleshoot code and concepts is encouraged. If you collaborate, list collaborators at the top of your submission.

All written responses must be written independently (in your own words).

Keep your work readable: Use clear headings and label plot elements thoughtfully (where applicable).

Submit both your rendered output and the Quarto ( `.qmd` ) file.

**Assignment submission**: Nathalie Bonnet

## Introduction

In this assignment, you will replicate the instrumental variable analysis from Stokes (2015), which examined how local wind turbine projects influenced electoral outcomes. Building on the matched dataset from Assignment 2, we will use Two-Stage Least Squares (2SLS) to estimate the causal effect of having a wind turbine proposed nearby on the change in Liberal Party vote share between 2007 and 2011. The instrument used in Stokes (2015) is a measure of local wind resource (average wind power, logged), which predicts where wind turbines are proposed. By using this instrument, we aim to isolate the portion of variation in turbine placement that is as-good-as-random, helping to meet the assumptions for causal identification.

**Study:** Stokes, 2015 – Article

**Data source:** Dataverse – Stokes, 2015 replication data

> Note: The estimates you obtain may not exactly match the published results in Stokes (2015) due to the alternative matching procedure used for processing the data in the previous assignment. Estimates should approximate the findings reported in *Table 2* of the article.

## Load packages

```
library(tidyverse)
library(janitor)
library(here)
```

```
library(jtools)   # for export_summs (pretty regression tables)
library(AER)      # for ivreg (2SLS estimation)
```

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## Load the matched dataset (from Assignment 2)

> The matched_data has been preprocessed by matching on key covariates (e.g. pretreatment home values, education, income, population density) to improve balance between treated and control precincts. We will now use this data for the IV analysis. Make sure to re-code the `precinct_id` variable as a `factor`.

```
# Load in matched data
matched_data <- read_csv(here("data", "matched_data_subset.csv"))
```

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## Part 1: IV Identification Rationale

Intuition for Using an Instrument:

**Question 1:** After matching on observables, why might we still need to utilize an instrumental variable approach to identify the causal effect of turbine proposals on vote share? In other words, what potential issues remain that an IV method can help address in this context? Use specific examples from the study to illustrate threats to a causal interpretation, then explain how an IV approach is designed to mitigate those threats.

*Response:* Even after matching, wind turbine project placement is non-random. A variable that would influence the outcomes through the presence of turbines is the geographic placement based on power generation/ wind speed. Locations with higher wind speeds are more likely to be selected as locations for projects, which influences the second stage (actual proposal or implementation) of turbines themselves, which in turn changes the outcome of treatment variables. An instrumental variable approach uses wind speed as a parameter to account for the presence or absence of turbines. This isolates the actual causal effect from the effect of the confounder, so that the estimate for the treatment effect on the outcome variable (liberal vote share) is more accurate.

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## Part 2: Two-Stage Least Squares (2SLS) Step-Wise Implementation

### 2A. First-Stage Estimation: Regress the treatment ($D$) on the instrument ($Z$)

$$D_i = \alpha_0 + \alpha_1 Z_i$$

a. Estimate the first-stage regression of the treatment on the instrument (with controls). Regress `proposed_turbine_3km` on `log_wind_power`.

b. Include the control variables used in Stokes (2015) for both stages: Distance to lakes, geographic coordinates (latitude & longitude) with their squares and interaction, plus district fixed effects.

c. After running the first stage, report the F-statistic for the instrument.

```
first_stage <- lm(proposed_turbine_3km ~ log_wind_power +
                      mindistlake + mindistlake_sq +
                      longitude + long_sq +
                      latitude + lat_sq +
                      long_lat +
                      factor(district_id),
                  data = matched_data)

export_summs(first_stage, digits = 3,
             model.names = c("First stage: Propoposed Turbine 3km"),
             coefs = c("(Intercept)", "log_wind_power"))
```

| | First stage: Propoposed Turbine 3km |
|---|---|
| (Intercept) | 15.027 |
| | (74.243) |
| log_wind_power | 0.711 *** |
| | (0.092) |
| N | 708 |
| R2 | 0.419 |

*** p < 0.001; ** p < 0.01; * p < 0.05.

## Testing Instrument Relevance

Check instrument strength (F-statistic)

```
# Extract the F-statistic for instrument strength
first_stage_summary <- summary(first_stage)

first_stage_summary$fstatistic[1]
```

```
   value
14.70538
```

**Question 2A:** Based on the instrument relevance test reported in the study, would you conclude the instrument is strong enough to be credible? Explain what a weak instrument would mean in this setting: Specifically, what would it suggest about compliance with Ontario's Green Energy Act policy?

*Response:* The F-statistic is reported as 68. As this is larger than 10, this suggests a that wind power is a strong instrument in the study. The calculated F-statistic is 14.7. Though comparatively smaller, both F-statistics are greater than 10, so we can make inferences from IV analysis using wind power.

A weak instrument in this case would suggest that IV estimates using wind power are unreliable to predict turbine placement. This could mean the Ontario Green Energy Act's implementation is not just being

shaped by wind power, and might relate more to other covariates.

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## 2B. Second Stage Estimation

### Regress the outcome ($Y$) on the fitted values from the 1st stage ($\hat{X}_i$)

$$Y_i = \beta_0 + \beta_1 \hat{D}_i + \epsilon_i$$

a. Now estimate the second stage of the 2SLS.

b. First, use the first-stage model to generate the predicted values of proposed_turbine_3km for each precinct (these are $\hat{D}_i$).

c. Add these predicted values as a new column in matched_data (e.g. proposed_turbine_3km_HAT).

d. Then, regress the outcome change_liberal (the change in Liberal vote share from 2007 to 2011) on the predicted treatment (proposed_turbine_3km_HAT), including the same controls and fixed effects as in the first stage.

e. Fill in the code for these steps below to obtain the second-stage regression results.

Save predicted values $\hat{X}_i$ from first stage

```
matched_data$proposed_turbine_3km_HAT <- predict(first_stage)
```

### Estimate the second-stage regression

$$LiberalVoteShare_i = \beta_0 + \beta_1 \widehat{ProposedTurbine}_i + ControlVariables... + \epsilon_i$$

```
second_stage <- lm(
    change_liberal ~ proposed_turbine_3km_HAT +
        mindistlake + mindistlake_sq +
        longitude + long_sq +
        latitude + lat_sq +
        long_lat +
        factor(district_id),
    data = matched_data
)

export_summs(
    second_stage,
    digits = 3,
    model.names = c("Second stage: Change in Liberal Vote Share"),
    coefs = c("(Intercept)", "proposed_turbine_3km_HAT")
)
```

| | Second stage: Change in Liberal Vote Share |
|---|---|
| (Intercept) | 16.966 |
| | (15.432) |
| proposed_turbine_3km_HAT | -0.065 * |
| | (0.027) |
| N | 708 |
| R2 | 0.586 |

*** p < 0.001; ** p < 0.01; * p < 0.05.

### Interpreting the 2SLS Estimate

**Question 2B:** Imagine you are explaining your 2SLS findings to a policymaker in Ontario. What does the estimated coefficient on `proposed_turbine_3km_HAT` imply about the electoral impact of a local wind turbine proposal (within 3 km) on liberal vote share?

*Response:* There is a 6.5% predicted decrease in liberal vote share in areas within 3 kilometers of a proposed wind turbine. The coefficient value is seen to be statistically significant, so it is unlikely that this result would occur randomly. Therefore this supports the author's idea that wind turbine proposals are linked to voting retaliation.

**Question 2C:** Explain what it means that IV identifies a LATE in the context of the wind-turbine voting study. What specific subset of observations does the second-stage 2SLS estimate apply to, and what does it imply about interpretation and generalizability?

*Response:* The IV identifies local average treatment effect (LATE) for the predicted treatment based on `proposed_turbine_3km_HAT` , not for `proposed_turbine_3km`. This means that the result is not generalizable. The first stage finds the effect of wind power on turbine proposal, while the second stage calculates the effect of turbine proposals on liberal vote share. The interpretation therefore relies on proposed turbines being influenced only by wind power.

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## Part 3: IV Assumptions and Validity

### Evaluate Instrument Validity

**Question 3:** List the four key assumptions required for the IV strategy (2SLS) to identify a causal effect, and briefly explain what each one means in the context of this study. (Hint: think about what conditions a valid instrument must satisfy (relevance, exclusion,...)

*Response:* 1. Relevance: Wind power must strongly predict turbine placement, so in this context high wind areas must be significantly more likely to have proposed turbines.

2. Exclusion Restriction: Wind speed itself doesn't influence voting.

3. Independence/Exogeneity: Wind power is uncorrelated with unobserved confounders.

4. Monotonicity: Higher wind speed doesn't make some precincts less likely to get turbines.

## Part 4: Estimate 2SLS using `AER::ivreg()`

📜 SEE Documentation for specification details: AER package Viggnette Example

> **Tip**
>
> Syntax for specifying 2SLS using `ivreg()`:
>
> `ivreg( Y ~ D + CONTROLS  | Z + CONTROLS , data )`
>
> - The first-stage predictor variables go after the `~` symbol
> - The second-stage predictor variables go after the `|` symbol

```
fit_2sls <- ivreg(change_liberal ~ proposed_turbine_3km +
                      mindistlake + mindistlake_sq +
                      longitude + long_sq +
                      latitude + lat_sq +
                      long_lat +
                      factor(district_id) |
                      log_wind_power +
                      mindistlake + mindistlake_sq +
                      longitude + long_sq +
                      latitude + lat_sq +
                      long_lat +
                      factor(district_id),
                  data = matched_data)

export_summs(fit_2sls, digits = 3,
             model.names = c("Change in Liberal Vote Share"),
             coefs = c("(Intercept)", "proposed_turbine_3km")
             )
```

| | Change in Liberal Vote Share |
|---|---|
| (Intercept) | 16.966 |
| | (15.737) |
| proposed_turbine_3km | -0.065 * |
| | (0.027) |
| nobs | 708 |
| r.squared | 0.570 |
| adj.r.squared | 0.549 |
| sigma | 0.082 |
| statistic | 27.814 |
| p.value | 0.000 |
| df | 34.000 |
| df.residual | 674.000 |

| | Change in Liberal Vote Share |
|---|---|
| nobs.1 | 708.000 |

*** p < 0.001; ** p < 0.01; * p < 0.05.

### Robustness checking strategies utilized in Stokes, 2015

**Question 4:** Choose two *robustness checks* from the paper that the authors use to increase confidence in their causal identification strategy. For each one, summarize the logic and findings from the robustness check in your own words:

*Response:* Instrumental Variable: The fixed effects model in this study is subject to selection bias in turbine placement driven by variation in unobserved exogenous factors. Isolating this variation based on wind power rather than political/social factors helps strengthen confidence in causal effect identification. The study finds similar results between methods, thus supporting causal interpretation for decreases in liberal vote share in precincts with proposed wind turbines.

Pre-Treatment Vote Share (2003): If areas with high wind happen to be less liberal, wind power should be a predictor of liberal vote share before turbines were proposed. Testing this relationship checks if political boundaries coincidentally align with wind patterns. The study did not find evidence to support this, the relationship between retaliatory voting and wind turbine proposals only emerges after proposals, which further supports causality.

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## END

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